



Defensive Security using Wazuh SIEM: Implementation, Log Monitoring, and Threat Detection

GROUP ASSIGNMENT REPORT

Bachelor of Information and Communication
Technology (BICT) Honours Degree CTNT 32051
– Cyber Security Laboratory

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1. Introduction to SIEM and Wazuh

Security Information and Event Management (SIEM) systems are used to collect, analyze, and monitor logs generated by operating systems and applications. SIEM helps in detecting security threats such as unauthorized access attempts, suspicious activities, and system misuse.

Wazuh is an open-source SIEM platform that provides log monitoring, intrusion detection, and realtime alerting. It is widely used for defensive security because it can detect attacks early and help administrators take corrective actions.

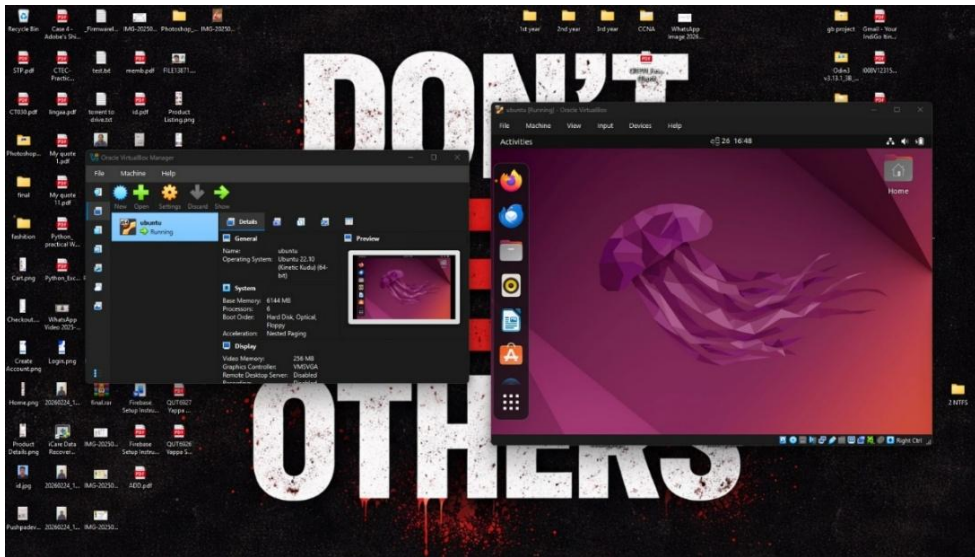
In this assignment, Wazuh SIEM was installed and configured to monitor a Linux system and detect security-related events.

2. System Architecture and Setup

2.1 Actual Environment Used in This Assignment

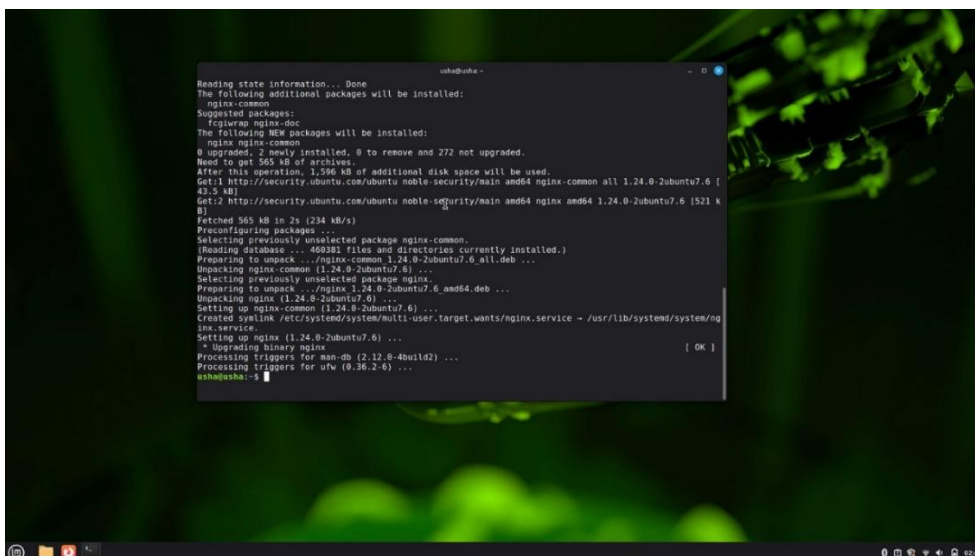
This assignment was implemented using two different systems, which makes it closer to a real-world environment.

- Ubuntu Virtual Machine
 - Runs Wazuh Server (Manager)
 - Runs Wazuh Indexer
 - Runs Wazuh Dashboard



(Wazuh Manager service running on Ubuntu VM)

- Linux Mint Laptop (Real Machine)
 - Runs Wazuh Agent
 - Generates system, authentication, and application logs



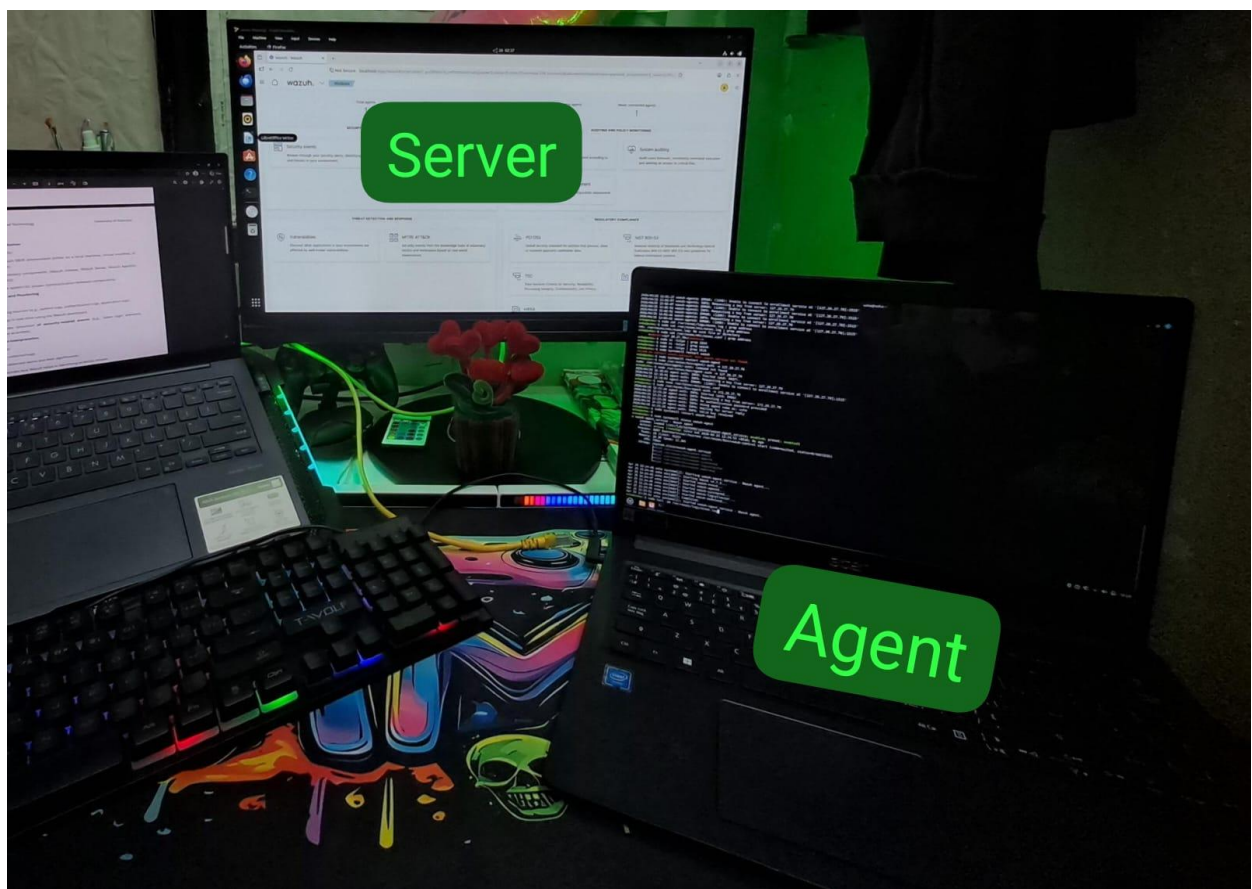
(Linux Mint desktop running on real laptop (Wazuh Agent))

2.2 Architecture Explanation

The Linux Mint system acts as the monitored endpoint. Logs generated on this system are collected by the Wazuh Agent. These logs are securely sent to the Wazuh Server running on the Ubuntu virtual machine. The server analyzes the logs, the indexer stores them, and the dashboard displays alerts and events.

Log flow:

- Linux Mint (Agent) → Ubuntu VM (Wazuh Server) → Indexer → Dashboard



(System Architecture)

3. Step-by-Step Installation Procedure

3.1 Wazuh Server Installation (Ubuntu Virtual Machine)

The following steps were performed on the **Ubuntu VM**

1. System update:

`sudo apt update && sudo apt upgrade -y`

2. Download Wazuh installation script:

`curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh`

3. Install Wazuh components:

`sudo bash wazuh-install.sh -a`

This installed:

- Wazuh Server (Manager)
- Wazuh Indexer
- Wazuh Dashboard

4. Service verification:

`sudo systemctl status wazuh-manager`

A screenshot of a terminal window on an Ubuntu system. The terminal shows the command `sudo systemctl status wazuh-dashboard` being executed. The output indicates that the `wazuh-dashboard.service` is loaded, active (running), and enabled. It was started on Sun 2026-04-26 at 01:23:06. The terminal also shows a series of log entries from the `opensearch-dashboards` service, indicating successful responses to various API requests. The terminal window has a dark background with a light-colored text cursor.

Wazuh Dashboard Running

All services were running successfully.

3.2 Wazuh Agent Installation (Linux Mint Laptop)

The following steps were performed on the Linux Mint system.

1. Install Wazuh Agent:

`sudo apt update`

`sudo apt install wazuh-agent -y`

2. Configure server address:

`sudo nano /var/ossec/etc/ossec.conf`

Server IP was set to Ubuntu VM IP.

3. Start and enable agent:

`sudo systemctl enable wazuh-agent`

`sudo systemctl start wazuh-agent`

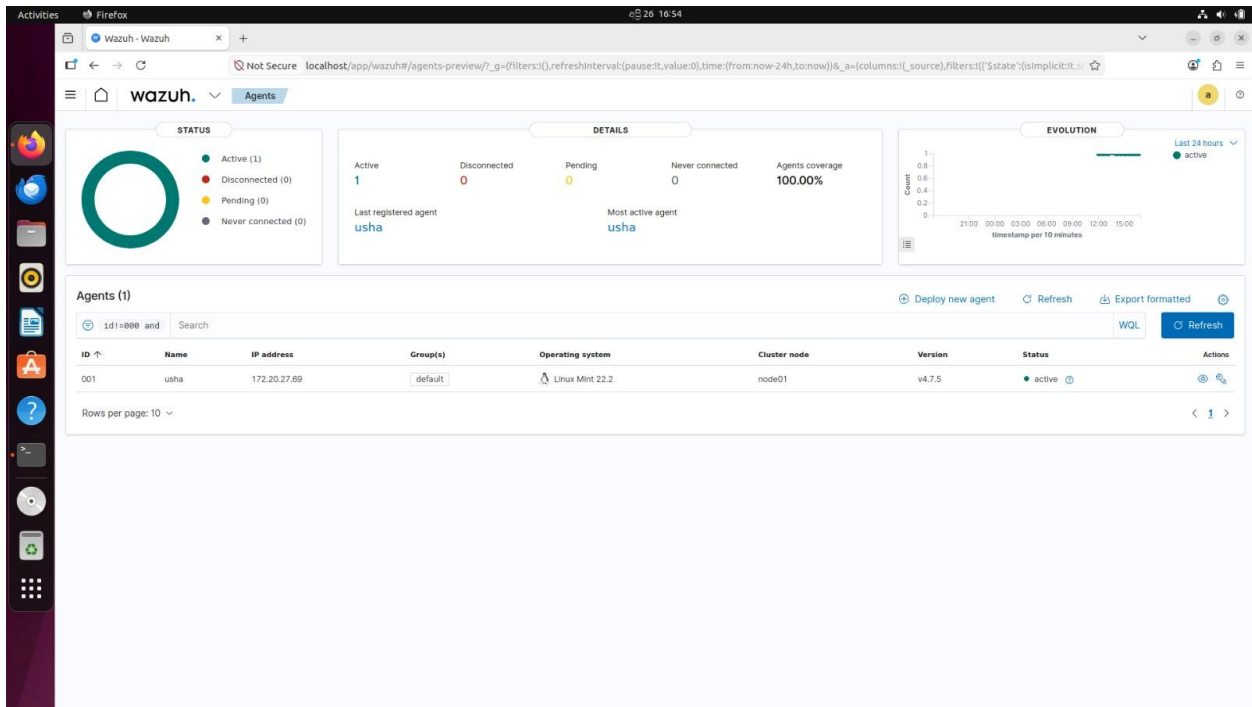
`sudo systemctl status wazuh-agent`

```
usha@usha:~$ sudo systemctl status wazuh-agent
wazuh-agent.service - Wazuh agent
Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: enabled)
Active: active (running) since Sat 2026-04-25 22:06:16 +0530; 4h 45min ago
Tasks: 32 (limit: 9133)
Memory: 17.3M (peak: 22.1M)
CPU: 1min 10.399s
CGroup: /system.slice/wazuh-agent.service
└─ 1346 /var/ossec/bin/wazuh-execd
   1357 /var/ossec/bin/wazuh-agentd
   1371 /var/ossec/bin/wazuh-syscheckd
   1385 /var/ossec/bin/wazuh-logcollector
   1402 /var/ossec/bin/wazuh-modulesd

Apr 25 22:06:08 usha systemd[1]: Starting wazuh-agent.service - Wazuh agent...
Apr 25 22:06:08 usha env[4324]: Starting Wazuh v4.7.5...
Apr 25 22:06:10 usha env[4324]: Started wazuh-execd...
Apr 25 22:06:11 usha env[4324]: Started wazuh-agentd...
Apr 25 22:06:12 usha env[4324]: Started wazuh-syscheckd...
Apr 25 22:06:13 usha env[4324]: Started wazuh-logcollector...
Apr 25 22:06:14 usha env[4324]: Started wazuh-modulesd...
Apr 25 22:06:16 usha env[4324]: Completed.
Apr 25 22:06:16 usha systemd[1]: Started wazuh-agent.service - Wazuh agent.
usha@usha:~$
```

(Agent Status)

4. Agent appeared as **Active** in the Wazuh Dashboard.



(active agent)

4. Configuration Details

4.1 Communication Between Server and Agent

Log communication uses encrypted TLS channels.

Linux Mint agent sends logs to Ubuntu VM server.

Connectivity was verified from dashboard → Endpoints → Agents.

4.2 Log Source Configuration (What Was REALLY Done)

All log configuration was done on **Linux Mint (agent machine)**.

Configuration file:

sudo nano /var/ossec/etc/ossec.conf

The following log sources were used:

- System Logs

<localfile>

<log_format>syslog</log_format>

<location>/var/log/syslog</location>

</localfile>

- Authentication Logs

<localfile>

<log_format>syslog</log_format>

<location>/var/log/auth.log</location>

</localfile>

- Application Logs

<localfile>

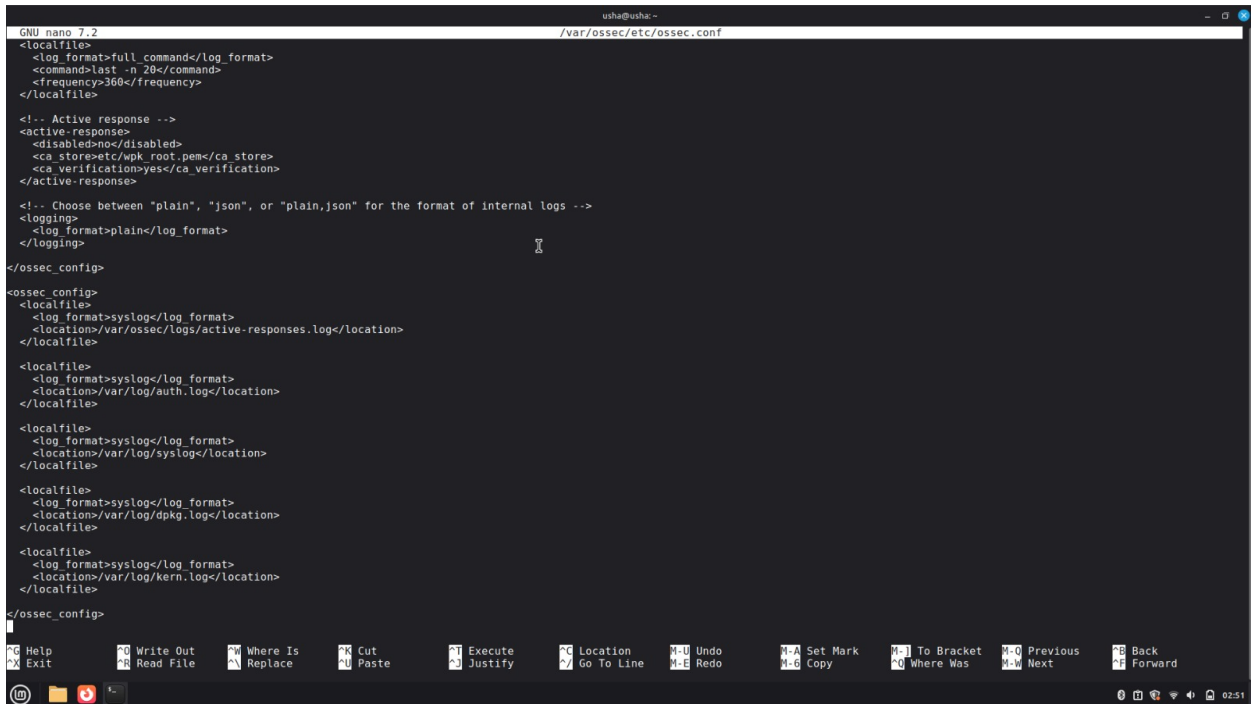
<log_format>syslog</log_format>

<location>/var/log/dpkg.log</location>

</localfile>

- Agent restart:

sudo systemctl restart wazuh-agent



```

GNU nano 7.2 usha@usha:~ /var/ossec/etc/ossec.conf
<localfile>
  <log_format>full_command</log_format>
  <command>last -n 20</command>
  <frequency>360</frequency>
</localfile>

<!-- Active response -->
<active_response>
  <disabled>no</disabled>
  <ca_store>etc/wpk_root.pem</ca_store>
  <ca_verification>yes</ca_verification>
</active_response>

<!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
<logging>
  <log_format>plain</log_format>
</logging>

</ossec_config>

<ossec_config>
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/auth.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/syslog</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/kern.log</location>
  </localfile>
</ossec_config>
  
```

Agent Configuration File (Linux Mint)

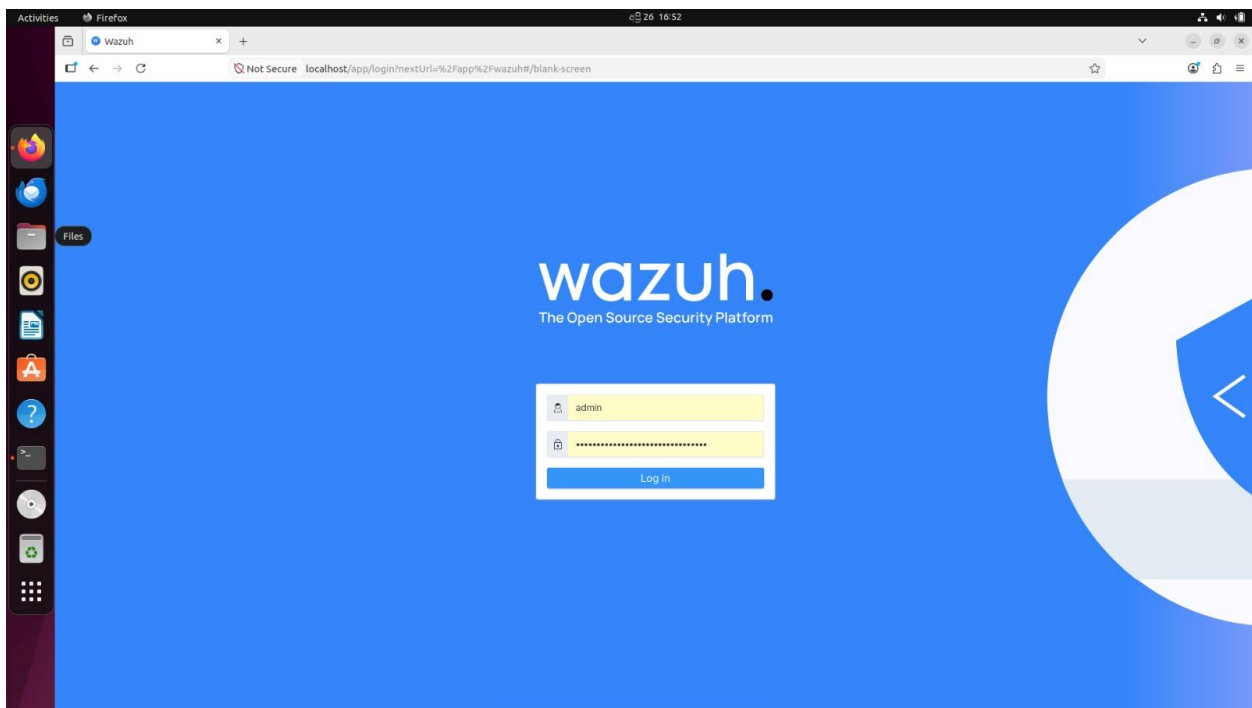
5. Log Monitoring Process

Log monitoring was done using the **Wazuh Dashboard on Ubuntu VM**.

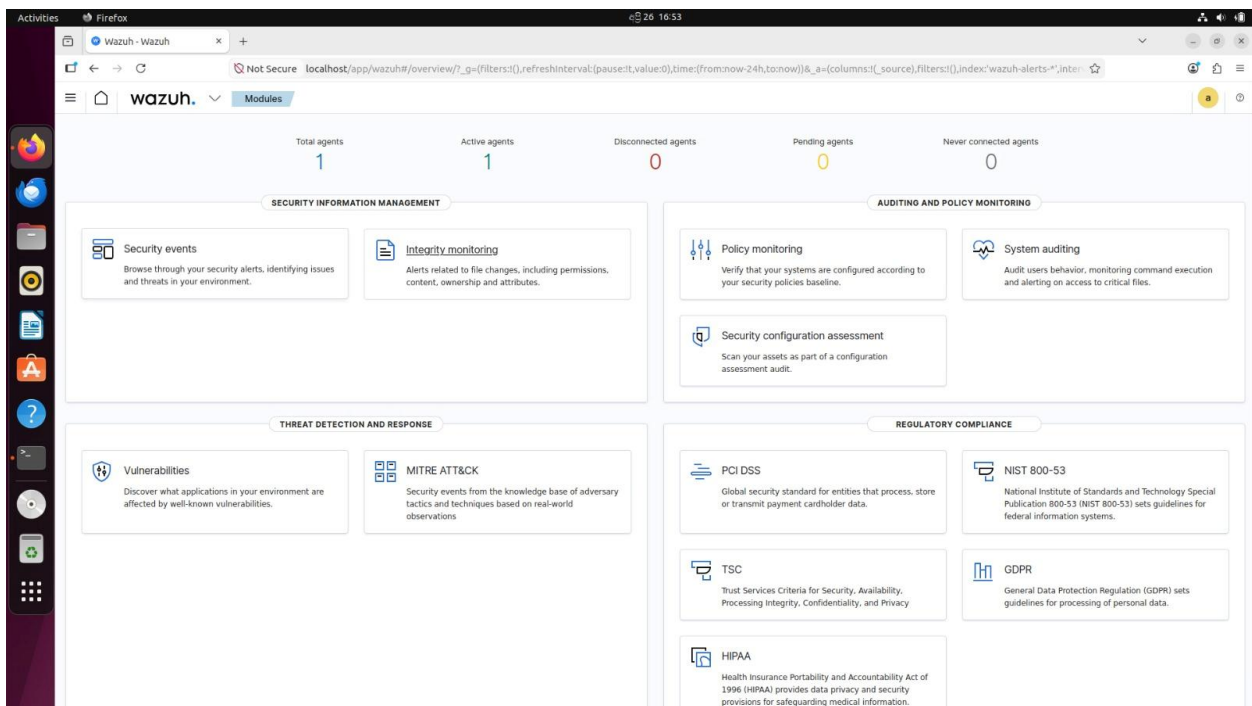
Steps followed:

1. Open dashboard in browser:

<https://<Ubuntu-VM-IP>>



Wazuh Dashboard Login Ubuntu VM



Wazuh Dashboard Main Home Page

2. Navigate to:

Security Events → Discover

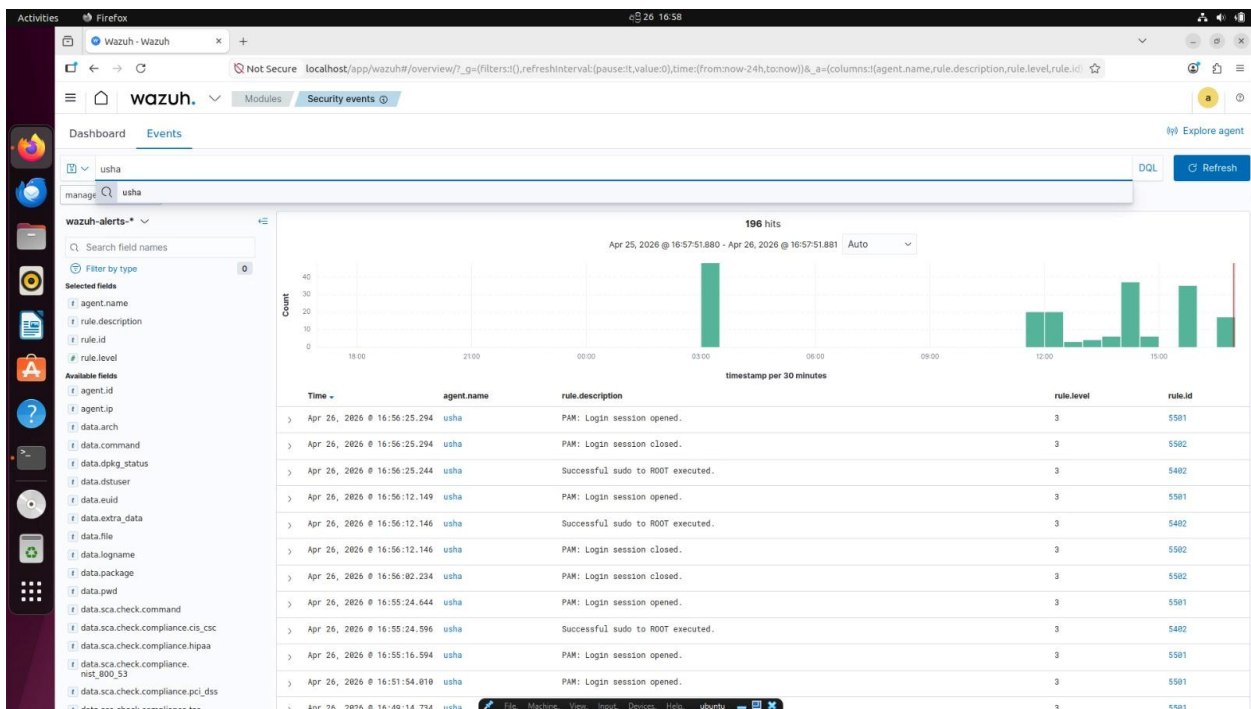
Security Events

3. Apply filter:

agent.name : linuxmint

rule.level >= 3

Logs were displayed and updated in real time as activities occurred on the Linux Mint system.

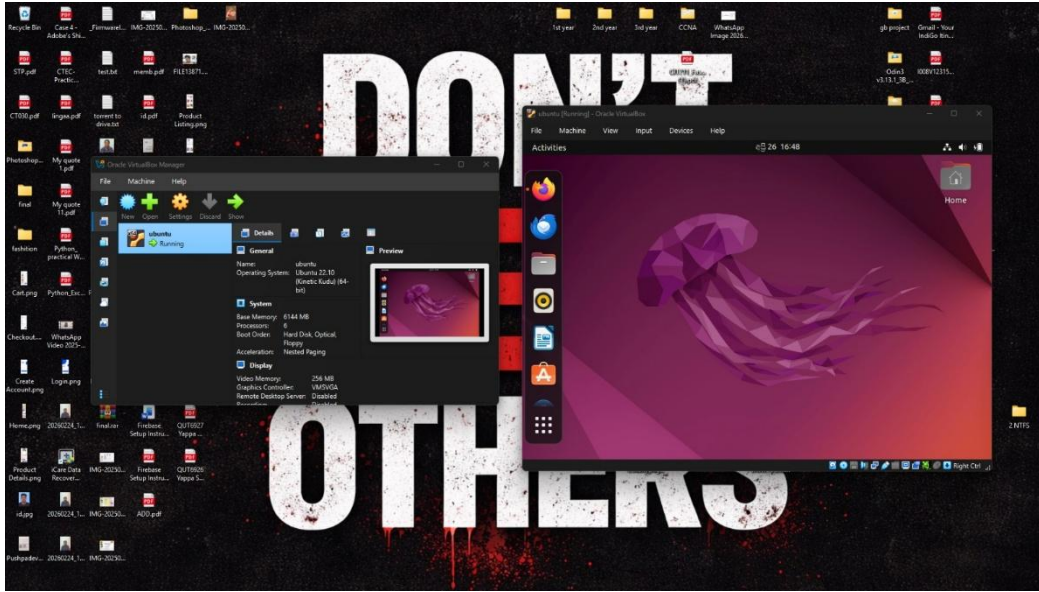


Log Monitoring

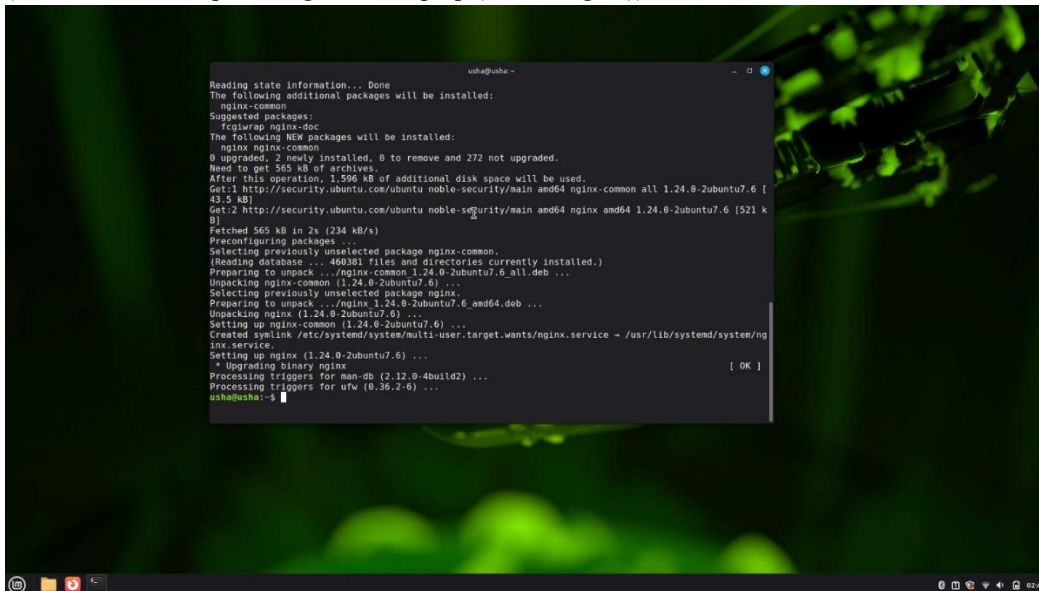
6. Screenshots of Implementation and Results

The following screenshots were captured:

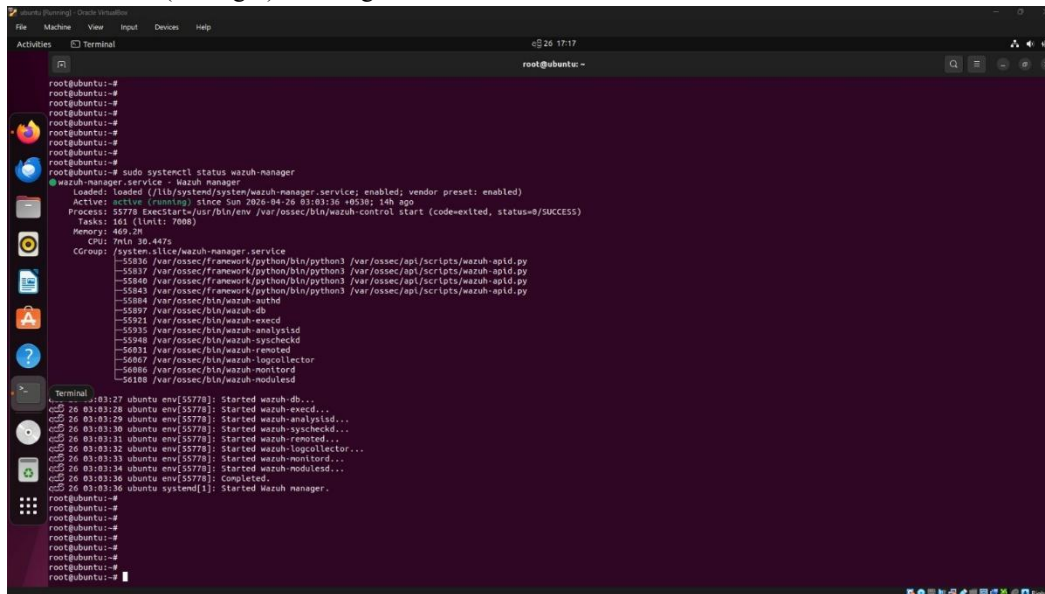
- (Wazuh Manager service running on Ubuntu VM)



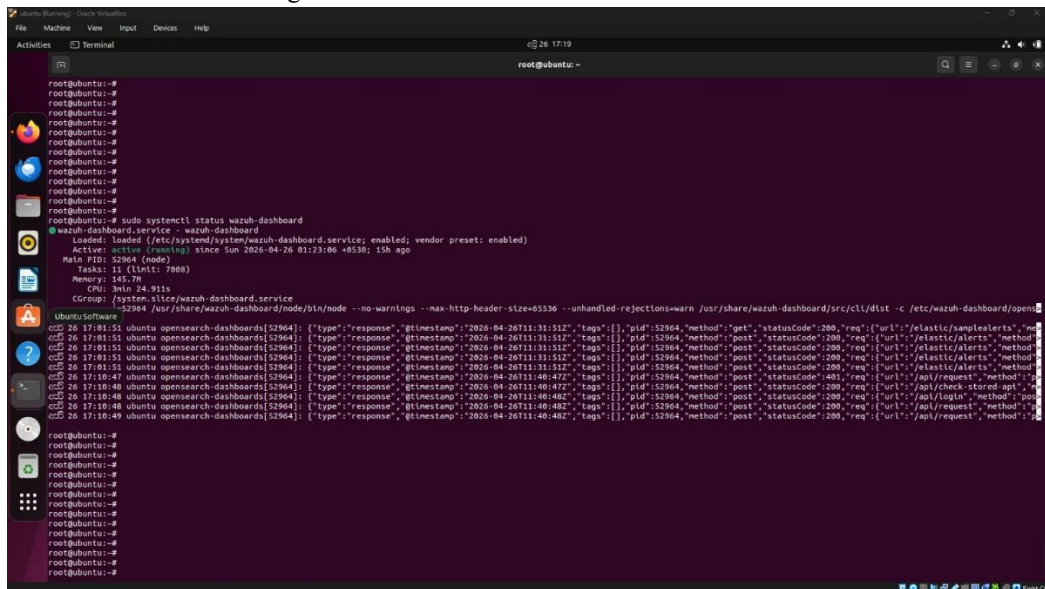
- (Linux Mint desktop running on real laptop (Wazuh Agent))



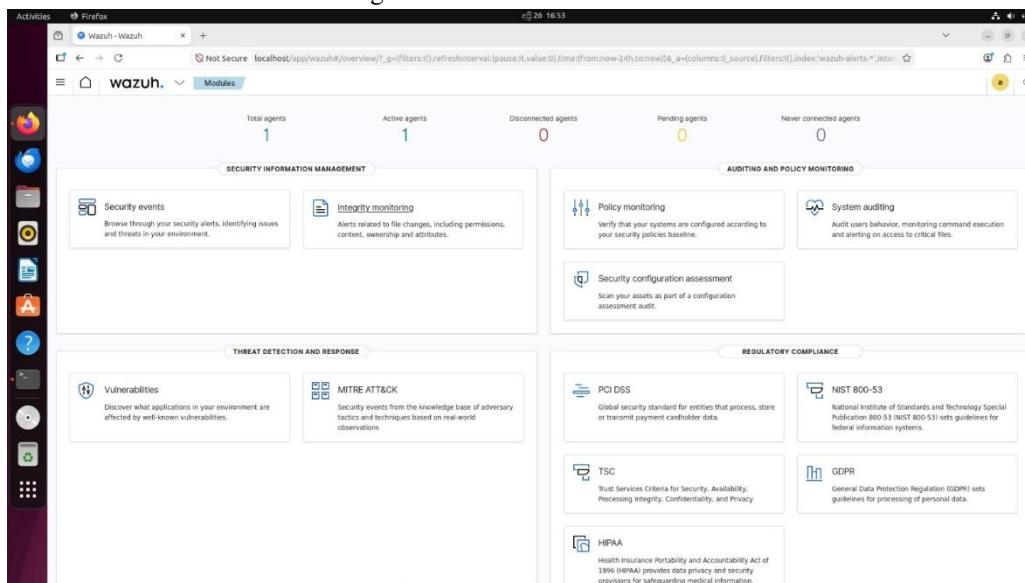
- Wazuh Server (Manager) Running



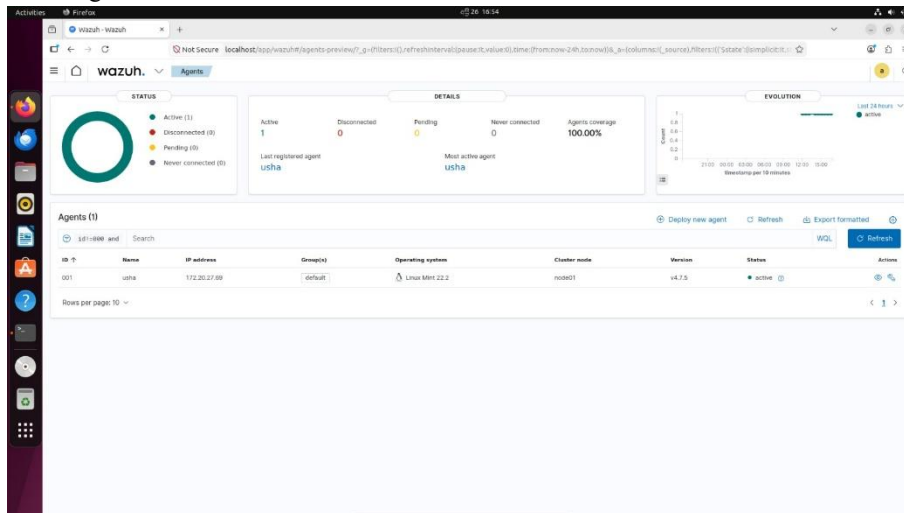
- Wazuh Dashboard Running



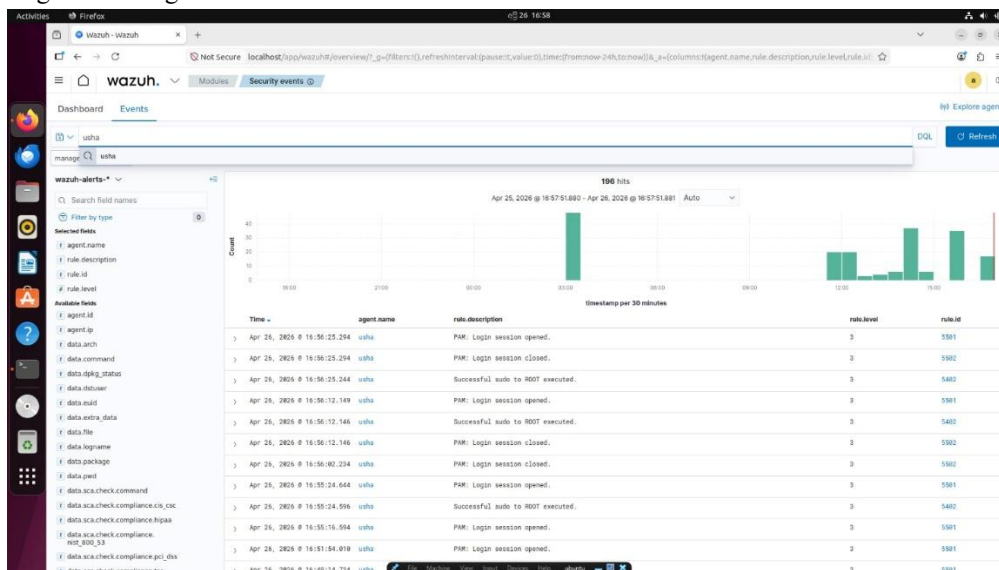
- Wazuh Dashboard Main Home Page



- active agent



- Log Monitoring

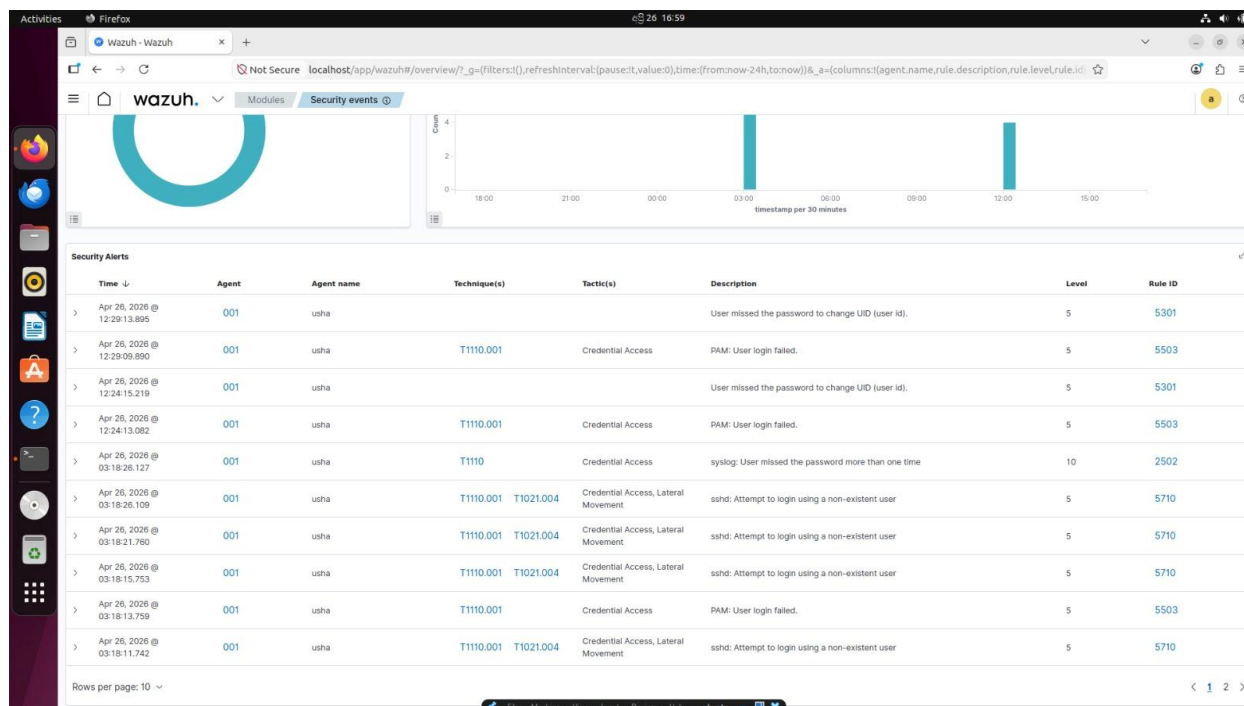


7. Analysis of Detected Events

7.1 Failed SSH Login Attempts

Command executed on Linux Mint:

ssh fakeuser@localhost



Failed Login Attempt Alert

Multiple incorrect passwords were entered.

What happened:

Wazuh detected repeated authentication failures from `/var/log/auth.log` and generated alerts.

Significance:

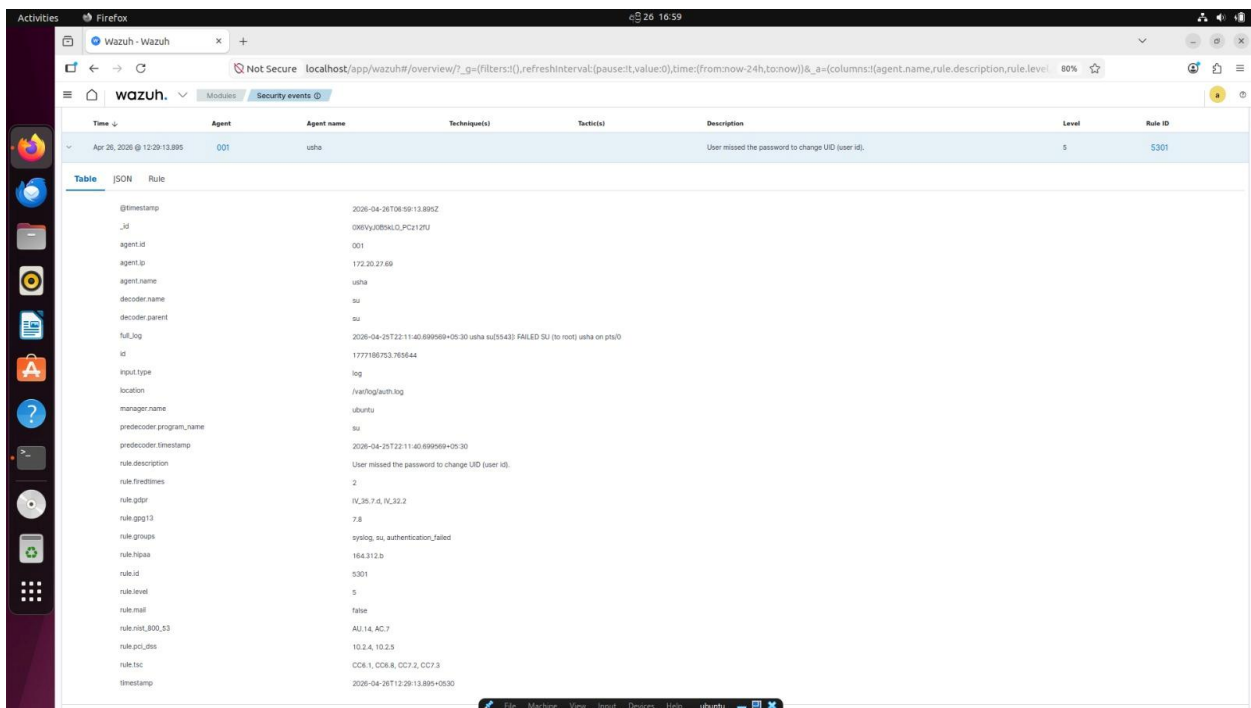
This indicates possible brute-force attempts to gain unauthorized access.

7.2 Unauthorized sudo Access

Command executed:

sudo su

Wrong password was entered.



Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Apr 26, 2026 @ 12:29:13.895	001	uaha			User missed the password to change UID (user id).	5	S301

Field	Value
@timestamp	2026-04-26T12:29:13.895Z
_id	0M6VvJ0B9Kd_PC212UJ
agent.id	001
agent.ip	172.30.27.69
agent.name	uaha
decoder.name	su
decoder.parent	su
full_log	2026-04-26T22:11:40.699569+05:30 uaha su(55433) FAILED SU (to root) uaha on pts/0
id	1777186753.765644
input.type	log
location	/var/log/auth.log
manager.name	ubuntu
predecoder.program.name	su
predecoder.timestamp	2026-04-26T22:11:40.699569+05:30
rule.description	User missed the password to change UID (user id).
rule.firetimes	2
rule.gidpr	16,35,7,6,16,32,2
rule.gidpr13	7,8
rule.groups	system, su, authentication_failed
rule.hipaa	164.312.0
rule.id	S301
rule.level	5
rule.mail	false
rule.nist_800_53	AU.14, AC.7
rule.pci_dss	10.2.4, 10.2.5
rule.tsc	CDK.1, CDK.8, CC7.2, CC7.3
timestamp	2026-04-26T12:29:13.895+05:30

Alert Details View (Auth log)

What happened:

Wazuh detected unauthorized privilege escalation attempts.

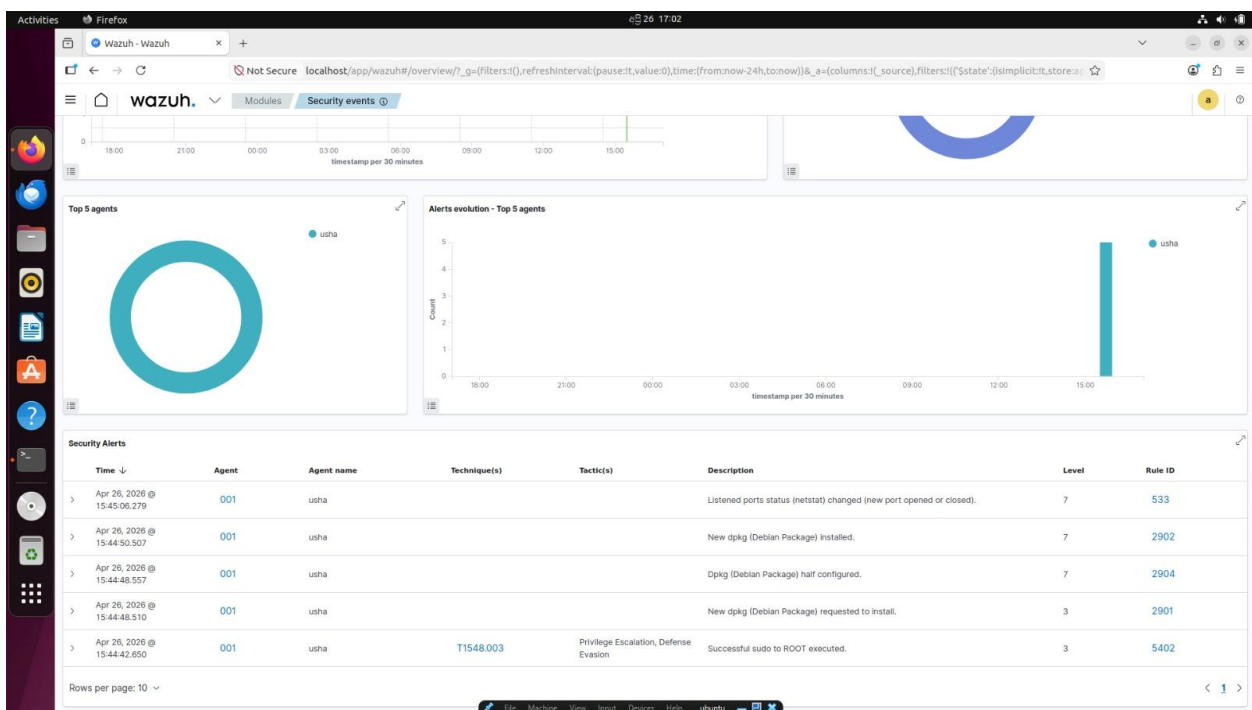
Significance:

Such attempts can lead to full system compromise if successful.

7.3 Application Activity Monitoring

When installing or updating packages:

sudo apt install <package-name>



Application Log Monitoring dpkg

The screenshot shows the 'Alert Details View' for the application log. It displays a table with various fields related to the alert, including timestamps, agent information, and specific log data.

Field	Value
@timestamp	2026-04-26T15:14:50.507Z
_id	cH9yZ085HLO_PC2tWgU
agent.id	001
agent.ip	172.20.27.69
agent.name	usha
data.arch	all
data.dpkg.status	status installed
data.package	nginx-common
data.version	1.24.0-2ubuntu7.6
decoder.name	dpkg-decoder
full_log	2026-04-26 01:36:11 status installed nginx-common all 1.24.0-2ubuntu7.6
id	1777198480.807249
input.type	log
location	/var/log/dpkg.log
manager.name	ubuntu
rule.description	New dpkg (Debian Package) installed.
rule.firetimes	1
rule.gidpr	TC26.7.d
rule.gsig13	4.10
rule.groups	syslog, dpkg, config_changed
rule.hipisa	164.312.5
rule.id	2902
rule.level	7
rule.mail	false
rule.mist_800_53	AU.8, AU.14
rule.pci_dss	10.6.1, 10.2.7
rule.tsc	CC1.2, CC3.3, CDE.8, CDE.1
timestamp	2026-04-26T15:44:50.507+0930

Alert Details View (application log)

What happened:

Wazuh monitored application-level changes through dpkg logs.

Significance:

Identifies unauthorized or suspicious software installations.

8. How Wazuh Helps Identify Threats

- Collects logs from multiple sources
- Uses predefined security rules
- Detects suspicious behavior in real time
- Displays alerts with severity levels
- Helps administrators respond quickly

9. Conclusion

This assignment successfully demonstrated the installation and use of Wazuh SIEM for defensive security. A realistic setup with a Wazuh Server on an Ubuntu virtual machine and a Wazuh Agent on a Linux Mint laptop was implemented. Log sources were configured, security events were generated, and alerts were analyzed. The results show that Wazuh is an effective tool for real-time monitoring and threat detection.

10. References

[1] Wazuh, "Wazuh Documentation," [Online]. Available: <https://documentation.wazuh.com/current/index.html>

[2] Canonical, "Ubuntu Documentation," [Online]. Available: <https://ubuntu.com/documentation>